

Chi Square vs F Tests for GLM Model Comparisons

Asked 5 years, 2 months ago Modified 5 years, 2 months ago Viewed 5k times



I've been creating some models in R using `glm()` and `rxglm()`. I'm experienced in building GLMs but my memory of some of the underlying theory is a little rusty.

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I'm interested in comparing model fits *for nested models* using chi-square tests, F tests, etc.



I'm able to compare nested glm model objects using



```
anova(model1, model2, test = "Chisq")
```

etc. From reading around the subject a little, it seems that chi-square is only valid for certain GLMs - those where the scale parameter is fixed (Poisson & binomial), whereas the F test should be used where the scale parameter is estimated (eg normal, gamma). Is this correct?

I have a particular interest in creating GLMs using the Tweedie family of distributions. Is this a case where F would be preferable to chi-square?

r

generalized-linear-model

chi-squared-test

f-test

tweedie-distribution

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edited May 28, 2019 at 10:13



kjetil b halvorsen ♦

81.5k 32 199 651

asked May 27, 2019 at 23:56



Alan

211 2 6

1 Answer

Sorted by: Highest score (default) ▾



Basically, yes. F is used when the dispersion parameter is estimated rather than assumed to be fixed to some known value. F is also often used for quasi-likelihood models where a somewhat *ad hoc* overdispersion parameter is estimated (see e.g. Venables and Ripley *Modern Applied Statistics with S*).

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For a Tweedie distribution you're estimating a dispersion parameter and (possibly) a shape parameter. If you estimate the shape parameter rather than fixing it some *a priori* value, you'll probably be underestimating the uncertainty unless you do something fancy (e.g. bootstrapping).



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answered May 28, 2019 at 2:21



Ben Bolker

44.7k 3 119 169

▲ Thank you. For the Tweedie application I'm using a fixed a priori shape parameter for the time being although I know that techniques exist for estimating it. I've done some `anova()` calls using `test = "F"` instead of `test = "Chisq"` and I'm getting identical p-values (for a number of different GLM error structures - Poisson, gaussian, gamma, tweedie). I didn't expect this - should I have? – Alan May 28, 2019 at 11:38

▲ Also for fitting GLMs I often have to use the `rxGlm()` function in the `revoScaleR` package because it performs so much better with large data sets. However the objects that the `rxGlm()` function returns don't seem to work with the `anova()` function, so that: `anova(rxModel1, rxModel2, test = "whatever")` and `anova(as.glm(rxModel1), as.glm(rxModel2), test = "whatever")` both fail. Is there a way around this, I wonder? Apologies - I haven't managed to format this comment properly (not sure how!) – Alan May 28, 2019 at 11:39 ✎

1 ▲ Huh. For your first question, I'm surprised too. Can you post a reproducible example? For your second question: there's an [unanswered question about this on SO](#) ... (oh, never mind, I see that it's you ...) I don't have Microsoft R installed and don't really plan to, so it's going to be hard for me to find a solution for you ... – Ben Bolker May 28, 2019 at 13:26 ✎
